



MIT SLOAN SCHOOL OF MANAGEMENT
MIT SCHWARZMAN COLLEGE OF COMPUTING



MODULE 1 UNIT 3
Video 1 Transcript

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PAUL MCDONAGH-SMITH: We've been building a vocabulary. We've defined agentic AI, positioned it on the AI continuum, and explored the architecture that turns a large language model into an agent. The tools, the planning, the memory, the feedback loops. We've examined the building blocks.

Now, I'd like to shift our lens because in practice, no organization deploys a building block. They deploy a workflow. And that distinction between understanding what an agent can do and understanding how an agent does real work is where the conversation changes from conceptual to operational. It's also frankly where a lot of organizations can get stuck.

So, let me frame what I mean by an agentic AI workflow, why it matters, and what questions you should be holding as we move into the next set of conversations.

What is an agentic AI workflow

MCDONAGH-SMITH: When most people first encounter generative AI, the interaction is transactional. You open a chat interface, you type a prompt, you get a response. Perhaps you refine it, iterate it a few times, and arrive at something useful. That's often a single prompt interaction and it's valuable, but it's also self-contained. The human is doing the orchestrating. The human decides what to ask next, when to switch tools, when to loop back, and when to check.

An agentic AI workflow is something fundamentally different. It's a structured sequence of tasks, a chain of reasoning, actions and decisions that an agent or a system of agents can execute with a degree of autonomy. Think of it as the difference between asking a colleague a question and giving that colleague a project brief.

With a question, you get an answer. With a brief, you get a process. Research, analysis, drafting, verification, delivery. The work unfolds across steps, across tools, and across time. And the value isn't in any single step. It's in the coherence of the sequence.

So, when we talk about agentic AI workflows, we're talking about AI systems that can manage that kind of end-to-end process, ingesting inputs from multiple sources, decomposing a complex objective into its subtasks, executing each step using the appropriate tool or data source, evaluating interim outputs and delivering a result. All with varying degrees of human oversight along the way.

A large proportion of the value that organizations are looking for from AI doesn't live in isolated tasks. It lives in the connective tissue between tasks, in the handoffs, in the sequencing, the judgment calls that link one step to the next.

Consider what happens when a customer files a complaint today. Someone reads it, someone categorizes it, someone pulls up the account history, someone checks the relevant policy, someone drafts a response, reviews it. Each of these steps might be simple, but the workflow, the choreography of all of these six steps is where the time goes, where error creeps in, and where the customer experience is won or lost.

An agentic workflow doesn't just speed up one of those steps, it reimagines the sequence itself. And that's a qualitatively different kind of value proposition than something along the lines of we made the drafting forty percent faster.

But reimagining a workflow isn't the same as automating it. Automation implies taking what exists and removing the human. Workflow redesign asks what I would argue is a more interesting question. If you were designing this process from scratch with agentic capabilities available to you today, what would it look like? Which steps really require human judgment? Which steps really benefit from machine speed and consistency? Where really should the agent act autonomously? And where should it pause and escalate? These are design questions and they are also leadership questions.

The translation problem

MCDONAGH-SMITH: Now, there is a significant gap between understanding agentic workflows conceptually and operationalizing them inside your organization. I'd like to name three dimensions of that gap because they're going to be central to the conversations ahead.

The first is the definitional challenge. The field is moving so fast that the very meaning of terms like agents and workflow is moving beneath our feet. Capabilities that required elaborate scaffolding a year ago are often now embedded into the models themselves. What counted as an agentic system six months ago may now look like a standard feature today. This creates a real coordination problem inside our organizations. If your engineering team, your product team, and your executive team each carry a different mental model of what an agent is, you're going to struggle to align on where and how to deploy one. Getting to a shared vocabulary isn't a semantic exercise. It's a strategic prerequisite.

Now the second is the organizational readiness question. Agentic workflows don't drop neatly into existing operating structures. They cross functional boundaries. They require data to flow between systems that may never have been connected. They demand new forms of collaboration, not just between people and machines, but between teams that have historically perhaps operated in silos. The technical challenges of building an agentic workflow are real. However, in my experience, they are rarely the binding constraint. The binding constraint is more often the organizational plumbing, the processes, the incentive structures, the cultural readiness to work differently.

And the third is the question of pace. How fast should you move? The instinct in many organizations is to study, evaluate, compare, pilot cautiously, and scale slowly. That instinct served us well in previous technology cycles, where the landscape was more stable and the cost of getting it wrong was high. But the landscape of generative AI and agentic systems isn't as stable. The models are improving on a cadence of weeks, not years. The tools and frameworks are proliferating. The competitive window is compressing. And so, one of the critical leadership questions is, what does responsible speed look like in this environment? Not recklessness but not caution either.

SPEAKER: Did you understand all the concepts covered in this video? If you'd like to review any of the content, please click on the chapter menu.